Group 4 SUSTAIN CDT Time Series Hackathon

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Data Quality & Integrity Issues

- **Severe Data Attrition:** The raw dataset contains 169,386 records, but only 44,135 rows survive the merging process—a massive $\sim$**74% data loss**.
- **Geographical Mismatches:** This attrition is primarily driven by uneven country coverage across the source files. Crop data covers 270 countries, rainfall covers 217, and temperature data covers only 137.
- **Temporal Gaps Root Cause:** The years **1988** and **2003** are completely absent from the final timeline because they are missing directly from the source `rainfall.csv` file.
- **Data Type Anomalies:** The *average_rain_fall* column introduces mixed data types (strings instead of floats) during the merge, requiring coercion/cleaning before modeling.
- **Validation Strategy:** Implemented a chronological split on the surviving rows:
  - **Training Set ($<$ 2009):** 35,749 rows (81% of data).
  - **Test/Validation Set ($\geq$ 2009):** 8,386 rows (19% of data).

Model Performance

- **R-Squared (R^2):** 0.8941
  - The model is highly explanatory, successfully capturing **89.4%** of the variance in crop yields across the validation set.
- **Mean Absolute Error (MAE):** 16,397 kg/ha
  - On average, our predictions deviate by roughly 16.4k kilograms per hectare from the true observed yields.
- **Baseline Context & Relative Error:**
  - The global average yield across the dataset is **74,706.46 kg/ha**.
  - Contextually, our MAE represents an average relative error of just $\sim$**22%** $(16,397/74,706)$, demonstrating highly robust predictive capability despite the severe historical data gaps.

Overcoming Attrition: Custom UK Data Pipelines

To resolve the severe data loss and geographical mismatches of the raw dataset, we built custom Python pipelines to source high-quality, localized UK data directly:

- **Agricultural Data Extraction:** Developed `get_yield.py` to parse official government spreadsheets (`AUK-chapter7.ods`). The script maps tables to 12 distinct crop categories (e.g., Wheat, Barley, Potatoes) and standardizes calculations to yield in kg/ha.
- **Meteorological Data Scraping:** Scraped decades of historical weather data across 36 Met Office stations, explicitly filtering out anomalous stations (e.g., Lerwick) and coercing missing data into safe `NaN` types.
- **Robust Aggregation:** Calculated true yearly averages for temperature, frost days, rainfall, and sun hours, automatically skipping missing values so the final averages are not distorted.

Pipeline Execution & Data Validation

Running our custom extraction scripts successfully bypassed the structural issues in the original dataset, generating a clean, validated baseline for our models.

Crop Yield Pipeline Output

```
$ python3 get_yield.py
Loading AUK-chapter7-20250710.ods...
Processing Table_7_1 -> Cereals
Processing Table_7_2 -> Wheat
Processing Table_7_3 -> Barley
Processing Table_7_4 -> Oats
Processing Table_7_5 -> Oilseed rape
Processing Table_7_6 -> Sugar beet
Processing Table_7_7 -> Protein crops
Processing Table_7_8 -> Green vegetables
```

Modeling: XGBoost on Cleaned UK Data

Dataset & Strategy

- **Shape:** 288 records, 8 features.
- **Inputs:** Crop type, Year, and 5 Met Office weather variables.
- **Validation:** Strict chronological split to prevent time-series data leakage:
  - **Train:** 172 rows ($\leq$ 2007)
  - **Test:** 116 rows ($\geq$ 2008)

Model Performance

Testing Metrics

- **R^2 Score: 0.7041**
- **MAE: 6,742 kg/ha**

Training logs showed steady RMSE reduction, converging optimally around ~250 boosting rounds without immediate overfitting.

XGBoost: Feature Importance Analysis

Feature	Importance
Crop (encoded)	87.49%
Year	4.72%
Frost Days (`af_days`)	2.06%
Sun Hours (`sun_hours`)	1.83%
Min Temp (`tmin_degC`)	1.69%
Rainfall (`rain_mm`)	1.22%
Max Temp (`tmax_degC`)	0.99%

Key Insights:

- **Biological Baseline:** `Crop` type dominates (∼87.5%). This is physically expected, as the base mass/yield of different crops (e.g., heavy Sugar Beet vs. light Linseed) varies massively.
- **Agri-Tech Progress:** The `Year` variable (4.7%) successfully captures historical macro-trends, such as improvements in farming practices and fertilizers.
- **Climatic Impact:** Weather conditions account for the remaining ∼7.8%, with frost days and sunlight driving more yield variance than raw rainfall.

It's On The Internet!

Figure 1: https://cropforcaster.vnet.tel/

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